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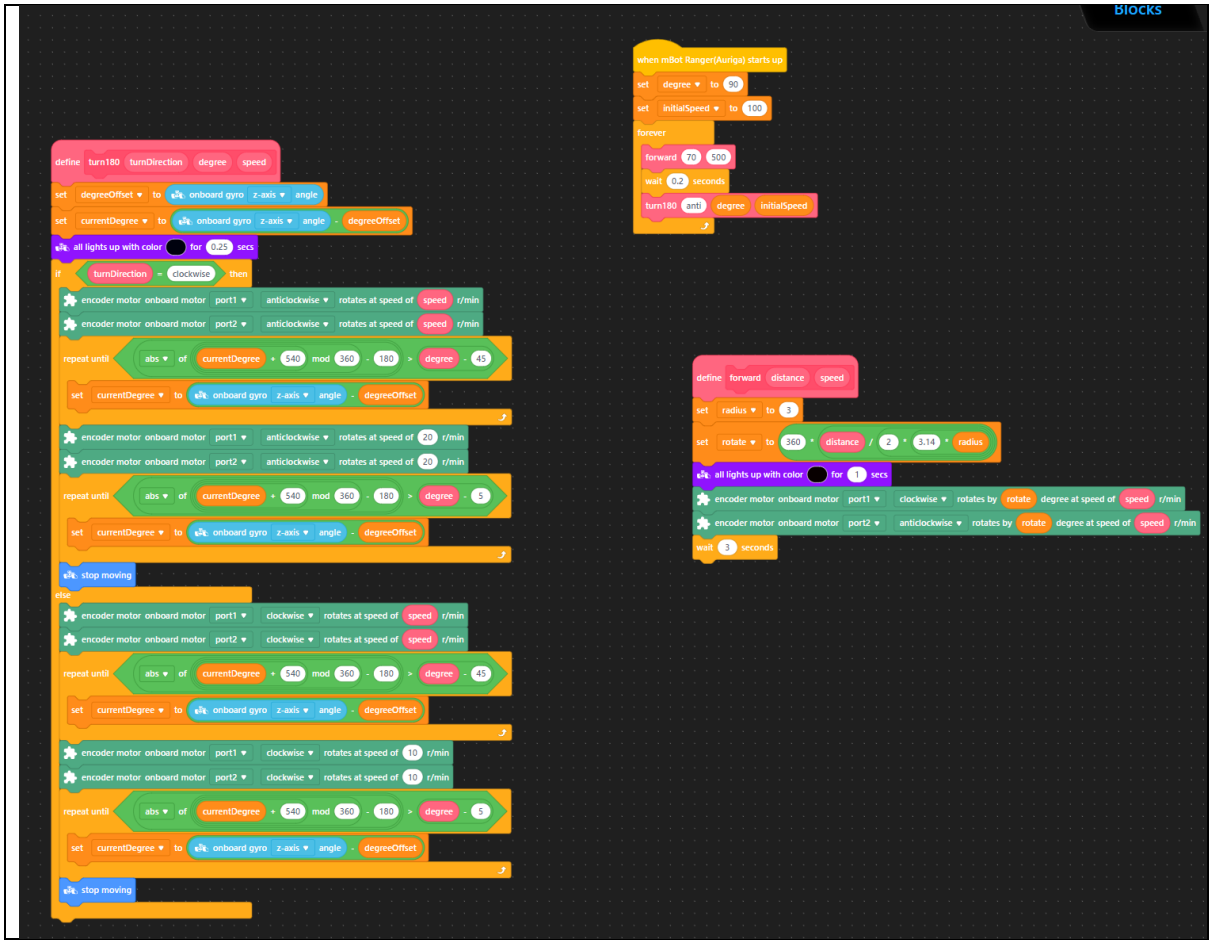
YouTube: rotation-100r/min rotation-500r/min square_1 square_2
GitHub: https://github.com/NoXxChapter/SIT122---Robotics-Studio.git

Which method did you use for straights? (Open Loop or Encoder)
Encoder

Compare the position errors from the Open Loop Square Test vs Encoder Motor Square Test vs Gyro-Enabled Square Test. You don't need to run Open Loop and Encoder Square Tests again – use the results from your previous task submission. Which one is the best? Why?
From my observed and experience hands-on. The Gyro-Enable gave a more accurate and smooth turn when compared to other two. I will not talk about Open Loop because it obviously the worst. But with remain two, it a difference story here. Encoder turn based on the calculate of geometric mathematic. By calculate the wheel's radius, distance between two wheels along with the turning degree's input. It gives us a general control of how to know when it turns to 90, 180, 360. But with Gyro-Enabled, it utilizes the hardware ability to know the coordinate of the current robot, by that we got a more accurate situation of robot at current to compare and calculate. Which result in far more accurate and impactful measurement.


Blocks
define turn180 turnDirection degree speed
set degreeOffset to onboard gyro z-axis angle
set currentDegree to onboard gyro z-axis angle - degreeOffset
all lights up with color for 0.25 secs
if turnDirection = clockwise then
encoder motor onboard motor port1 anticlockwise rotates at speed of speed r/min
encoder motor onboard motor port2 anticlockwise rotates at speed of speed r/min
repeat until abs of currentDegree + 540 mod 360 - 180 > degree - 45
set currentDegree to onboard gyro z-axis angle - degreeOffset
encoder motor onboard motor port1 anticlockwise rotates at speed of 20 r/min
encoder motor onboard motor port2 anticlockwise rotates at speed of 20 r/min
repeat until abs of currentDegree + 540 mod 360 - 180 > degree - 5
set currentDegree to onboard gyro z-axis angle - degreeOffset
stop moving
else
encoder motor onboard motor port1 clockwise rotates at speed of speed r/min
encoder motor onboard motor port2 clockwise rotates at speed of speed r/min
repeat until abs of currentDegree + 540 mod 360 - 180 > degree - 45
set currentDegree to onboard gyro z-axis angle - degreeOffset
encoder motor onboard motor port1 clockwise rotates at speed of 10 r/min
encoder motor onboard motor port2 clockwise rotates at speed of 10 r/min
repeat until abs of currentDegree + 540 mod 360 - 180 > degree - 5
set currentDegree to onboard gyro z-axis angle - degreeOffset
stop moving
when mBot Ranger(Auriga) starts up
set degree to 90
set initialSpeed to 100
forever
forward 70 500
wait 0.2 seconds
turn180 anti degree initialSpeed
define forward distance speed
set radius to 3
set rotate to 360 * distance / 2 * 3.14 * radius
all lights up with color for 1 secs
encoder motor onboard motor port1 clockwise rotates by rotate degree at speed of speed r/min
encoder motor onboard motor port2 anticlockwise rotates by rotate degree at speed of speed r/min
wait 3 seconds